

Course Code: CS2005

Name of the Programme: B.Tech (Hons.) CSE

Name of the Course: Agile Software Engineering and Devops

Semester: IV

Course Credit:	No. of hours per week (L+T+P):	Total no. of Teaching hours:
3	2+0+2	60

Course Objectives

- Introduce the principles of Agile software development and compare them with traditional software development models.
- Explore various Agile methodologies such as Scrum, XP, Lean, and Kanban, and provide guidelines based on project needs.
- Develop the skills needed to gather software requirements, design specifications, test plans, and coverage analysis in Agile-based projects
- Equip students with hands-on experience in using modern DevOps tools (e.g., Git, GitHub, Docker) for automation, and deployment in Agile environments.

Syllabus

Module - 1 : Introduction to Agile	No. of hours
	5
Introduction to Software engineering (SDLC), Software process models- Waterfall, V model, Iterative model, Spiral model; Introduction to Agile, Agile Manifesto, Agile development approach, traditional development models, benefits and pitfalls of Agile development, Introduction to OOAD and UML.	

Module - 2 : Agile Methodologies and Practices	No. of hours
	5
Introduction to various Agile methodologies such as Scrum, XP, Lean, and Kanban, Scrum Roles - Product Owner, Scrum Master, Team, Release Manager, Project Manager, Product Manager, Architect, events, and artifacts; Product Inception: Product vision, stakeholders, initial backlog creation; Agile Requirements – User personas, story mapping, user stories, 3Cs, INVEST, Acceptance criteria, sprints, requirements, product backlog and backlog grooming; Test First Development; Pair Programming and Code reviews; code coverage and tools.	

Module - 3 : Sprint Management and Planning Techniques	No. of hours
	5
Sprint Planning, Sprint Reviews, Sprint Retrospectives Sprint Planning - Agile release and iteration (sprint) planning, Develop Epics and Stories, Estimating Stories, Prioritizing Stories (WSJF technique from SAFe), Iterations/Sprints Overview. Velocity Determination, Iteration Planning Meeting, Iteration, Planning Guidelines, Development, Testing, Daily Stand-up Meetings, Progress Tracking, Velocity Tracking, Monitoring and Controlling: Burn down Charts, Inspect & Adapt (Fishbone Model), Agile Release Train.	

Module - 4 :DevOps Foundations	No. of hours
	10
Overview of Devops and why it is needed; Continuous Integration, Build, Testing, Delivery and Deployment(CI and CD): Jenkins, Pylint, PyUnit. Overview of version control, branching and release using GIT;Git/Github/GitActions, Creating pipelines, Setting up runners Containers and container orchestration (Docker, DockerHub and Kubernetes) for application development and deployment.	

Module - 5 :Agile Testing and Automation	No. of hours
	5
Testing: Functionality Testing, UI Testing(Junit, Sonar), Performance Testing, Security Testing, Load Testing, Stress Testing, A/B testing; Agile Testing: Principles of agile testers; The agile testing quadrants, Agile automation, Test automation pyramid; Test Automation Tools: Selenium, RPA(UiPAth), Traceability matrix.	

Course Outcomes: After completing the course, the students will be able to	
a.	Evaluate the advantages and disadvantages of Agile development compared to traditional models.
b.	Assess various Agile methodologies such as Scrum, XP, Lean, and Kanban, and determine their appropriate applications.
c.	Create software requirements, design specifications, test plan and Analyze test coverage, requirements traceability for a software project.
d.	Utilize and implement various DevOps tools (e.g., Git, GitHub, Docker) in a software project.

Text Books:	
1	K. S. Rubin, <i>Essential Scrum: A Practical Guide to the Most Popular Agile Process</i> . Boston, MA: Addison-Wesley Professional, 2012. ISBN: 978-9332546172.
2	R. S. Pressman and B. Maxim, <i>Software Engineering: A Practitioner's Approach</i> , 9th ed. New York, NY: McGraw-Hill Higher Education, 2019. ISBN: 978-1259872976.
3	M. Cohn, <i>Succeeding with Agile: Software Development Using Scrum</i> . New Delhi, India: Pearson Education India, 2015. ISBN: 978-9332547964.
4	G. Kim, P. Debois, J. Willis, J. Humble, and J. Allspaw, <i>The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations</i> . Portland, OR: IT Revolution Press, 2016. ISBN: 978-1942788003.

Reference Books:

1	I. Sommerville, <i>Software Engineering</i> , 10th ed. Boston, MA: Pearson, 2015. ISBN: 978-0133943030.
2	A. Cockburn, <i>Agile Software Development: The Cooperative Game</i> , 2nd ed. Boston, MA: Addison-Wesley Professional, 2006. ISBN: 978-0321482754.
3	H. Kniberg, <i>Scrum and XP from the Trenches</i> , 2nd ed. [Place of publication not specified]: C4Media, 2015. ISBN: 978-1329224278.
4	J. Highsmith, <i>Agile Project Management: Creating Innovative Products</i> , 2nd ed. Boston, MA: Addison-Wesley Professional, 2009. ISBN: 978-0321658395.
5	J. A. Crowder and S. Friess, <i>Agile Project Management: Managing for Success</i> . Cham, Switzerland: Springer, 2014. ISBN: 978-3319090177.
6	A. Stellman and J. Greene, <i>Learning Agile: Understanding Scrum, XP, Lean, and Kanban</i> . Sebastopol, CA: O'Reilly Media, 2015. ISBN: 978-1449331924.
7	S. Vadapalli, <i>DevOps: Continuous Delivery, Integration, and Deployment with DevOps</i> . Birmingham, U.K.: Packt Publishing, 2018. ISBN: 978-1789132991.

Lab Programs /Tutorials		[30 Hours]
1	To understand the Waterfall model and its phases, and to analyze its advantages and disadvantages	
2	To understand the Agile Scrum methodology and its core principles, and to practice implementing it	
3	Identify functional and non-functional requirements of a real-time software project	
4	To understand the importance of software requirements specification and to practice creating a software requirements specification document	
5	To understand the importance of system modeling and to practice creating a system model using UML	
6	Write 2-5 test cases each testing category for a software project 1) Sanity Testing 2) Security Testing 3) Stress testing 4) Performance Testing	
7	Document the test plan, define list of test cases and traceability matrix of your respective team project using standard IEEE templates	
8	Identify your project's scrum team, Define Product Backlog, High Level Sprint Plan (consisting of max of 6 sprints of duration 1 week each)	
9	Identify your project's scrum team, Define Product Backlog, High Level Sprint Plan (consisting of max of 6 sprints of duration 1 week each)	
10	Live simulation of Daily Scrum, Sprint Planning, Sprint Review Sprint Retrospective meetings	
11	Hands on lab on Git, Github, and GitAction	
12	Hands on lab on Docker & Docker Hub	
13	To understand the importance of testing in software engineering and to practice testing a software project using PyUnit	
14	Create Agile Testing Quadrants for a real-time software project by putting 2-5 tests in each quadrant	
15	For Live team project do the end2end devops demo, prepare project report	